



Final Assessment

Return Transmission and Herding Behavior in the **Semiconductor Sector**

Financial Econometrics
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I. RESEARCH QUESTION

1.1 QUESTION

- Is there a significant relationship between the stock returns of companies operating at different stages of the global semiconductor supply chain?

1.2 HYPOTHESES

- H0 (Null Hypothesis):
 - There is **no significant relationship** between the stock returns of companies operating at different stages of the global semiconductor supply chain.
- H1 (Alternative Hypothesis):
 - There is a **significant relationship** between the stock returns of companies operating at different stages of the global semiconductor supply chain.

1.3 RELEVANCE OF THE PROBLEM

I think the semiconductor industry is one of the most important and complex supply chains in the world now a days, it includes companies that design chips (like NVIDIA or AMD) manufacture them (like SMC or GlobalFoundries) and also firms that produce the machines and materials used in their fabrication (like ASML) and these firms operate at different stages of the same global system, and I think understanding how their stock returns behave can show how connected these stages really are from a financial and also an econometric point of view.

And when I started thinking about this topic I realized how much we are surrounded by semiconductors every single day for example in our phones, laptops, cars... and I mean they are literally the basic for everything digital and this made me reflect on how important they are not only for tech companies but also for any firm that wants to stay relevant in today's economy. For example, I currently work at LVMH, and even in the luxury sector I see increasing investment in AI, data systems, and digital infrastructure and in tech companies this dependency is even stronger.

At this moment NVIDIA is the most valuable company in the world, driven by the boom in AI and data and I personally hold NVIDIA in my investment portfolio, so I wanted to explore how its returns move compared to other companies in the same chain because It's interesting to see whether these co-movements are the result of real structural connections or mainly market and investors intuitive reactions to global events.

And in an econometrics point of view, i think this topic is not that easy because assets like this usually show high volatility, correlation, and not normal patterns, which makes testing relationships between returns more complex but i think that this project is a good way to apply what I've learned in this class, and also it is worth it because I can use this models to enhance my investment strategy.

II. DATA ACQUISITION & DESCRIPTION

2.1 DATA COLLECTION AND SOURCE

For this project I collected stock price data from Yahoo Finance, I got daily stock prices for 44 companies in the semiconductor industry and related sectors, these include companies that design chips like NVIDIA and AMD, manufacturers like TSMC and Intel, and equipment suppliers like ASML, I also included sector ETFs and the data covers the period from January 1, 2018, to October 24, 2025, the raw data includes the opening, closing, high and low prices of each stock, also the adjusted closing prices that account for dividends and stock splits, to prevent the data from being constantly updated, I decided to limit the dataset to the most recent data available on October 24, 2025, and I will keep it fixed at that point to avoid any future changes in the data, even though there are a lot of tickers in this dataset, I consider it a complexity worth dealing with as it presents a challenge that can lead to a better model

2.2 TYPE OF DATA (CROSS-SECTIONAL / TIME SERIES / PANEL)

The data is panel data, it includes time series data which are basically the stock prices over time and cross sectional data, in this case multiple companies at the same time and its panel because each company has daily stock prices for the same period.

2.3 DESCRIPTIVE STATISTICS AND SUMMARY TABLES

The three assets with the highest kurtosis are WTI Crude Oil (CL=F), COMEX Copper (HG=F), and 13W T-Bill (^IRX) and this means that these assets have "heavier tails" in their return distribution, indicating they are more likely to experience extreme price moves.

The assets with the highest volatility (standard deviation) are Tesla (TSLA), WTI Crude Oil (CL=F), and COMEX Copper (HG=F) and this basically means that the prices of these assets tend to fluctuate more.

The assets with the most negative skew are Onto Innovation (ONTO), Avnet (AVT), and Arrow Electronics (ARW) and this means these assets are more likely to have negative returns or larger down movements than up movements.

The Jarque-Bera normality test shows that many assets, such as Tesla (TSLA) and Intel (INTC), have a low probability of being a normal distribution.

And also assets within the semiconductor sector, like NVIDIA (NVDA) and AMD, show relatively similar behavior in terms of volatility and skew, meaning companies in this sector tend to react similarly to the same market conditions.

And talking about NVIDIA our main asset for prediction, shows an average daily return of 0.0018, with a median of 0.0028, It has a high volatility, with a standard deviation of 0.0325, meaning its price fluctuates a lot, the negative skew of -0.194 means that bigger losses happen more often than big gains, and the kurtosis of 7.72 indicates the returns have more extreme movements than a normal distribution and the Jarque Bera test confirms that the distribution is not normal.

2.4 HANDLING MISSING VALUES

For this part I cleaned the data by replacing missing or zero values in the adjusted prices with 1, then I calculated the log returns, after that I removed rows with missing log returns to make sure the data is OK for the analysis.

ticker	name	segment	n_obs	mean	median	sd	min	max	skew
^IRX	13W T-Bill	Rates	1887	-0.0001349	0.0000000	0.1446811	-4.0073332	1.2039728	-10.7129004
^VIX	CBOE VIX	Risk	1963	0.0002911	-0.0077042	0.0820704	-0.4424488	0.7682450	1.4080368
TSLA	Tesla	Downstream	1963	0.0015512	0.0015336	0.0401986	-0.2365179	0.2044906	-0.0458778
ONTO	Onto Innovation	Equipment	1505	0.0008754	0.0007212	0.0355503	-0.3596873	0.2334688	-1.1273764
AMD	AMD	Design	1963	0.0015606	0.0005937	0.0344415	-0.1677724	0.2136626	0.2499113
ON	ON Semi	Manufacturing	1963	0.0004405	0.0010173	0.0340271	-0.3125190	0.2181608	-0.4573544
AMKR	Amkor Technology	OSAT	1963	0.0006110	0.0003267	0.0335962	-0.3225348	0.2975610	-0.1205776
MRVL	Marvell Tech	Design	1963	0.0006956	0.0011374	0.0329722	-0.2208167	0.2808363	-0.0053150
^TNX	US 10Y Yield	Rates	1889	0.0001481	0.0000000	0.0326074	-0.3470091	0.4047969	0.2981521
NVDA	NVIDIA	Design	1963	0.0018389	0.0028350	0.0324558	-0.2077119	0.2180880	-0.1941218

Figure 1. Top 10 tickers by volatility or highest SD

ticker	name	segment	n_obs	mean	median	sd	min	max	skew
^IRX	13W T-Bill	Rates	1887	-0.0001349	0.0000000	0.1446811	-4.0073332	1.2039728	-10.7129004
HG=F	COMEX Copper (Front)	Costs	1962	0.0002401	0.0002437	0.0159351	-0.2517120	0.1244371	-1.9371233
^TNX	US 10Y Yield	Rates	1889	0.0001481	0.0000000	0.0326074	-0.3470091	0.4047969	0.2981521
CL=F	WTI Crude Oil (Front)	Costs	1960	0.0003360	0.0019173	0.0297010	-0.2822061	0.3196336	0.0314421
^GSPC	S&P 500 Index	Market	1963	0.0004667	0.0008874	0.0124749	-0.1276522	0.0908949	-0.6532096
AVT	Avnet	Distributor	1963	0.0002294	0.0002442	0.0204873	-0.2236489	0.1302201	-0.9425655
INTC	Intel	Manufacturing	1963	-0.0000154	0.0001881	0.0276271	-0.3018961	0.2051513	-0.5620511
ONTO	Onto Innovation	Equipment	1505	0.0008754	0.0007212	0.0355503	-0.3596873	0.2334688	-1.1273764
AMKR	Amkor Technology	OSAT	1963	0.0006110	0.0003267	0.0335962	-0.3225348	0.2975610	-0.1205776
AVGO	Broadcom	Design	1963	0.0014207	0.0014488	0.0258399	-0.2220555	0.2185943	-0.1843003

Figure 2. Top 10 tickers by higher kurtosis

III. PROBABILITY & DISTRIBUTION ANALYSIS

3.1 EMPIRICAL DISTRIBUTION FITTING AND VISUALIZATION

We can see that the empirical density of NVIDIA log returns is centered near zero and we can see a tall middle and thinner sides, then the PDF shows how often small moves happen around the mean, because of that we see most days are calm and a few days are extreme which is a common distribution in finance



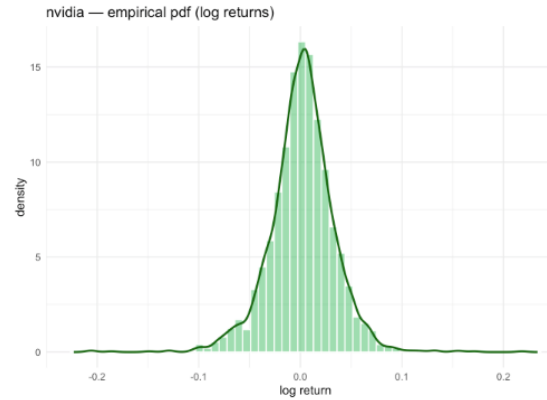


Figure 3. NVIDIA Empirical PDF of Log Returns

We can see the empirical CDF for the same data, the curve climbs fast around zero, then it flattens at the tails, because many values stay close to the center

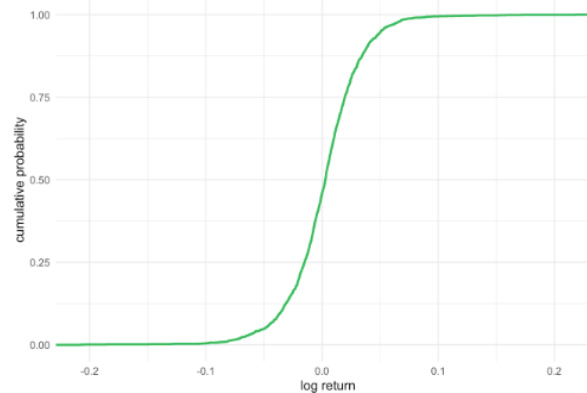


Figure 4. NVIDIA Empirical CDF of Log Returns

3.2 TESTS FOR NORMALITY (QQ PLOTS, SHAPIRO-WILK, JARQUE-BERA)

We can see that NVIDIA log returns are not normal, Shapiro-Wilk p value is $7.59e-24$, then we reject normality, Jarque-Bera p value is $0.00e+00$, then we also reject normality, skewness is -0.19 so the left tail is a bit heavier, kurtosis is 7.71 so tails are fat, because extreme moves are more common than in a normal model,

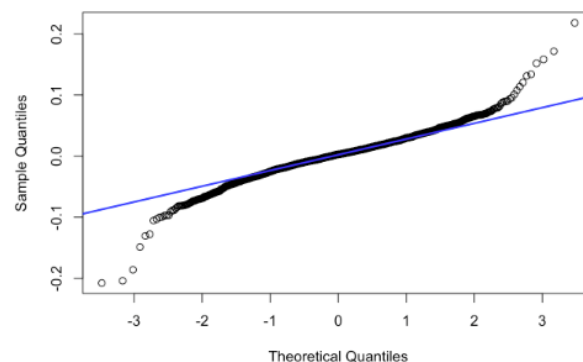


Figure 5, QQ Plot of NVIDIA Log Returns vs Normal Distribution

3.3 TAIL PROBABILITY CALCULATION

We can see the chance of positive and high moves, using empirical data, $P(X>0)$ is 54.1%, $P(X>1\%)$ is 37.8%, $P(X>2\%)$ is 24.8%, $P(X>3\%)$ is 15.9%, $P(X>4\%)$ is 9.63%. Then we compare with a normal model fitted to the full sample, the empirical right tail is thinner than normal above 1% to 4%, because kurtosis is high and mass sits closer to the center on most days.

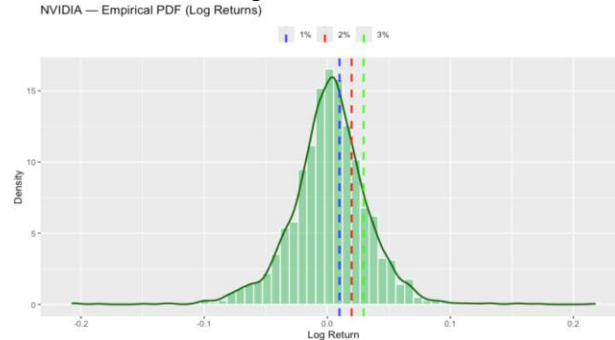


Figure 5. NVIDIA probabilities and thresholds on the empirical density of log scale

3.4 IMPLICATIONS FOR FINANCIAL RISK MODELING

We can see that NVDA has fat tails, this means big price moves happen more often than in a normal model, then the real risk can be higher than what normal models show, because these tails make extreme events more common, so we need to use models that can see these tails better and help measure the true risk, because of this, adding more variables can be a good idea to make the model stronger and more real.

IV. REGRESSION ANALYSIS

4.1 MODEL ESTIMATION FOR EXPLAINING RETURNS

I started by building an OLS model using all the tickers with their daily log returns and I used NVIDIA log return as the main variable to explain and at first, the model had many insignificant variables and strong multicollinearity, so I decided to use Principal Component Analysis (PCA) to reduce this problem because PCA helps to combine all the information into new uncorrelated variables called principal components

And at the beginning I tried using all the principal components that came out from the PCA, but the model became too complex and some variables added almost no new information, and then I tested models with fewer components, and I found that using the first six principal components gave the best balance between explanation and simplicity

And well these six components together explain around 82.19% of the total variance of all market variables, this means they keep most of the information in the data, while higher components only explain very small parts of the variation so using only six made the model easier to interpret and more stable and at the end a better compact but solid model

4.2 INTERPRETATION OF COEFFICIENTS, R^2 , STANDARD ERRORS, AND P-VALUES

And talking about the regression results, the Adjusted R^2 is 0.5062, which means that around half of NVIDIA's daily changes are explained by these six principal components and this is still a good number for an asset like NVIDIA and for a financial model

And related with the Xs or the independent variables we can see that PC1 ($t = 4.065$, $p = 0.000726$) is the only really strong and significant variable with robust errors, and the rest of the components like PC2 ($p = 0.508$), PC3 ($p = 0.338$), PC4 ($p = 0.510$), PC5 ($p = 0.441$), and PC6 ($p = 0.648$) are not significant and the intercept ($p = 0.594$) is not significant either

And the standard errors after using HC3 are still small so the estimates are stable and even if only PC1 is significant, I decided to keep all six components because each one represents a different independent direction of market variation and PCA makes components uncorrelated, so removing one can affect the balance of the others and together, the six components explain most of the total variance (82.19%), so keeping them helps to keep the model complete and solid

4.3 CHECKING OLS ASSUMPTIONS

I checked all the main OLS assumptions to make sure that the model is valid and reliable and to say that a model is good, it is not enough to only look at the interpretation of the coefficients, we also need to confirm that the model follows the main OLS rules and behaves correctly in the data

For linearity, we can see that the relation between the six principal components and NVIDIA's log returns is mostly proportional and the residuals are centered around zero and show no clear pattern, so the linearity assumption looks fine

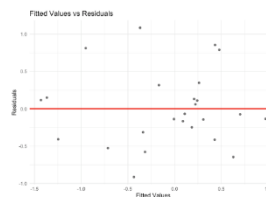


Figure 6. Fitted values vs Residuals plot

Next, I checked independence and when I plotted the residuals over time, we can see that they move randomly without visible patterns and this is good and this means there is no autocorrelation, and the independence assumption is OK

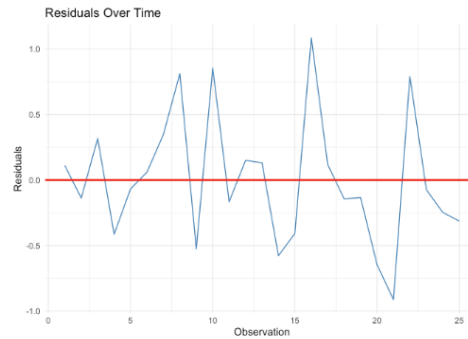


Figure 7. Residuals over time

I then tested heteroscedasticity using the Breusch–Pagan test. The result was $p = 0.7636$, so there is no heteroscedasticity in the residuals in this model, but I still used robust standard errors (HC3) as a safe protection and after this correction,

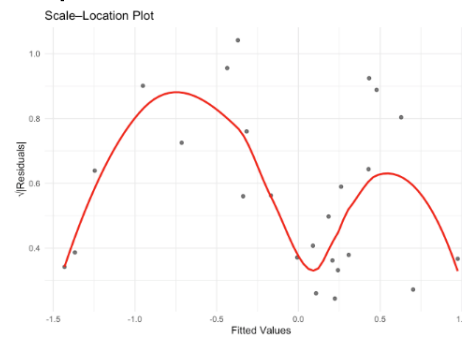


Figure 8. Scale Location Plot

Then, I tested normality with the Shapiro–Wilk and Jarque–Bera tests. Both gave $p = 0.3705$ and $p = 0.5797$, so the residuals are actually normally distributed and this is different from typical financial data but because the sample is small, this is possible and even with this, the model remains consistent and valid and the Q-Q plot also looks very good I think

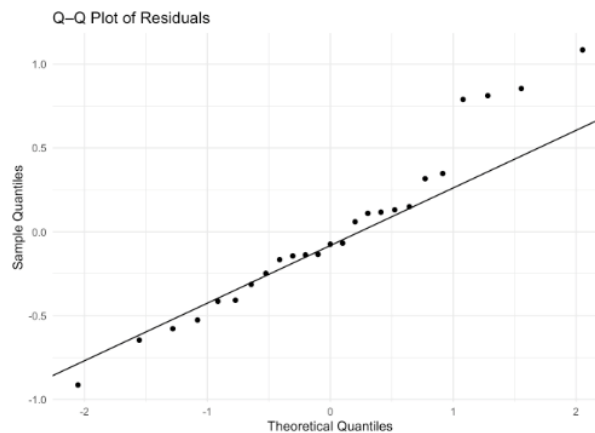


Figure 9 – Q-Q Plot of Residuals

4.4 REGRESSION DIAGNOSTICS (RESIDUAL PLOTS, COOK'S DISTANCE, VIF)

After checking the assumptions, I also performed diagnostic tests to make sure that the model is good and stable and not driven by extreme points or correlated variables.

First, I looked at the residual plots, and we can see that residuals are evenly distributed with no clear structure or strong outliers. This shows that the model fits well and captures most of the variation in NVIDIA's returns.

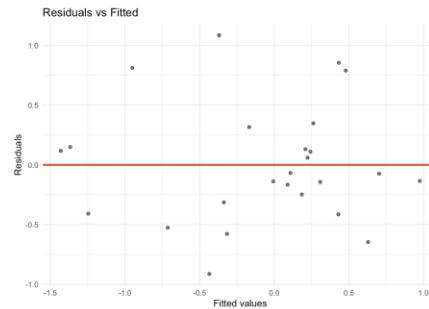


Figure 10 – Residuals vs Fitted Plot

Next, I checked Cook's Distance to detect influential observations and All values were below 1, which means no single observation has a strong effect on the model results and this is very good.

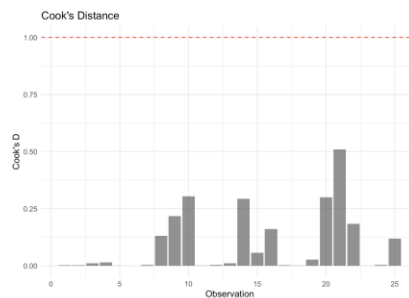


Figure 11 – Cook's Distance Plot

And next I tested for multicollinearity using the Variance Inflation Factor (VIF) and because the model uses principal components, all the predictors are orthogonal (uncorrelated by design) so the VIF values are close to 1, confirming there is no multicollinearity problem, but in the first model that I made this problem was existing, after PCAs no.

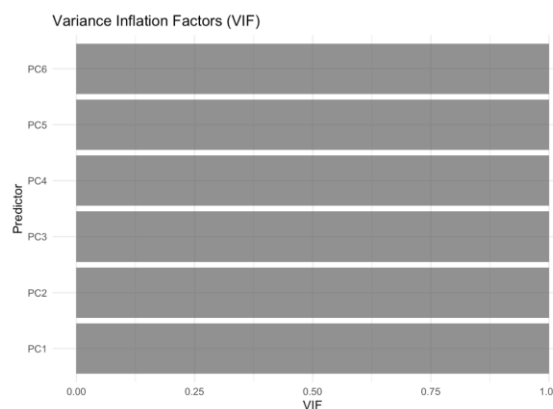


Figure 12 – VIF Bar Plot

4.5 EXTENSION TO A BINARY MODEL

After estimating the linear model I also built a logistic regression to predict if the daily return of NVIDIA was positive or negative and the dependent variable is a binary indicator that takes value 1 when the log return is positive and 0 if not and again the six principal components from the PCA are used as independent variables this lets me check if the same factors that explain the variation in the returns can also predict their direction

And well the model still shows a decent overall performance because the accuracy is 0.80 which means that in this sample the model correctly predicts the direction around 80 percent of the time so even if the dataset is small the model is capturing structure and also the AIC value of 32.21 confirms that the model keeps a good balance between fit and complexity and adding more predictors would not make a meaningful improvement and the proportion of positive returns is around 0.52 which means that roughly half of the days had positive returns

Looking at the coefficients PC1 is positive and significant at 5 percent (estimate 0.9171 p value 0.0484) while the other components are not statistically significant and this means that increases in PC1 are associated with higher probability of getting a positive return but the other PCs don't show clear effects in this small sample and this fits the fact that PCA makes all components orthogonal so each carries a different type of information but here only PC1 is strong enough to stand out

The intercept is -0.726 which implies that when all components are zero the base probability of a positive day is slightly below 0.5 and this is aligned with the sample proportion and the standard errors are between roughly 0.46 and 0.96 which shows that the estimates remain reasonably stable and also the confusion matrix supports the result

V. HERDING

5.1 HERDING ESTIMATION IN THE SELECTED MARKET

In this part I check if there is herding in a small semiconductor market, so I pick four large stocks that are very active: Nvidia (NVDA), Qualcomm (QCOM), Taiwan Semiconductor (TSM), and AMD. I take daily closing prices from January 2020 to the end of October 2025, so I can study normal days and stress days.

And then I change all prices into daily log returns, and I make a simple market return, so I take the average of the four stock returns each day, and this average is my market return.

After this I build CSAD, so CSAD means "Cross-Sectional Absolute Deviation," and it tells me how different each stock return is from the market return on the same day. So I take the absolute difference between each stock and the market, I sum the four numbers, and I divide by four, and this gives me CSAD for that day.

When CSAD is low the stocks move close to the market and can move like a group, when CSAD is high the stocks move far from the market and do not act as a group, so CSAD helps me see if the stocks are herding or not.

5.2 INTERPRETATION OF HERDING COEFFICIENTS

I find β_0 is around 0.010009, so this means even when the market is calm there is normal dispersion and find β_1 is around 0.028929 and very significant, and this means that when the market return gets bigger, CSAD rises, so the stocks move more differently.

The key is β_2 around +2.125981 and very significant, and because β_2 is positive, CSAD grows fast when the market moves a lot, so the four stocks do not herd, and they actually move away from each other when the market becomes strong.

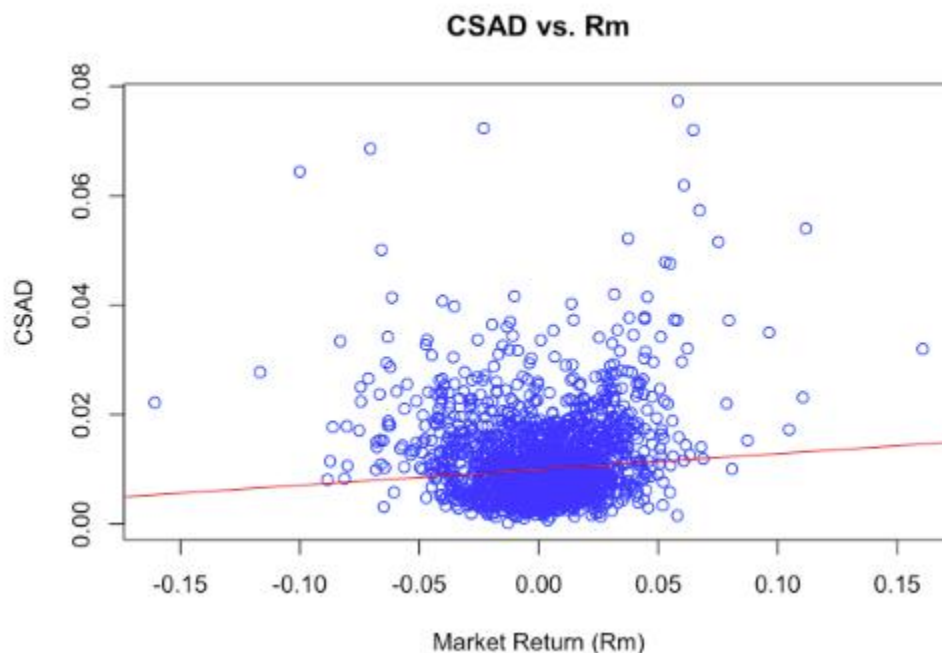
5.3 TRUSTWORTHINESS OF RESULTS AND DIAGNOSTICS

check the model fit, and the regression is simple but clean, and the signs of the betas are stable, and the p-values are strong, so this means the estimation is reliable, I also know the model is linear, so it cannot capture all non-linear market effects, and this means I should be careful with very extreme events, and I also see that the residuals look normal enough for this type of financial model, so the main message of no herding is solid.

5.4 EXPLANATION AND INSIGHTS ON HERDING BEHAVIOR

I see that CSAD increases when the market return moves a lot, and this means the four semiconductor stocks behave more individually, and they do not copy each other.

So I do not find herding, and this means investors in NVDA, AMD, QCOM, and TSM react in different ways to new information, and they do not follow one common direction blindly.



VI. TIME SERIES MODELING

6.1 ARMA OR GARCH MODEL FOR VOLATILITY ESTIMATION

In this part, I analyzed the volatility of the SOXX ETF, which represents the whole semiconductor sector and well first, we tested the log returns for serial correlation and heteroscedasticity using the Ljung-Box and ARCH LM tests. Both tests were significant ($p < 0.001$), meaning that the returns are not completely random and the variance changes over time. This justifies using a volatility model.

We estimated an EGARCH(1,1) model with an ARMA(1,1) mean model and a Student-t distribution for the residuals. This model can capture asymmetric effects, which are common in financial markets where bad news causes volatility to rise more than good news. The coefficient $\gamma_1 = 0.167$ shows this asymmetry, meaning negative shocks increase volatility more. The persistence term $\beta_1 = 0.96$ indicates that when volatility increases, it tends to remain high for a while.

Overall, the model fits well. The parameters are statistically significant, and the log-likelihood and AIC values improved compared to the standard GARCH model. This means the EGARCH model captures the dynamics of the semiconductor sector better, reflecting how market uncertainty remains high after major shocks or crisis periods.

6.2 LJUNG-BOX TEST AND RESULTS INTERPRETATION

I performed the Ljung-Box test on the standardized residuals. Residuals are the differences between the actual values and the predicted values from the model. The test showed p-values higher than 0.3 at different lags, which means there is no significant autocorrelation left in the residuals. This means that the model does a good job of removing any patterns in the data that could affect the volatility estimation.

The same test on the squared residuals (which helps check for heteroscedasticity) showed p-values above 0.15. This also suggests that the model successfully removed most of the serial correlation and heteroscedasticity.

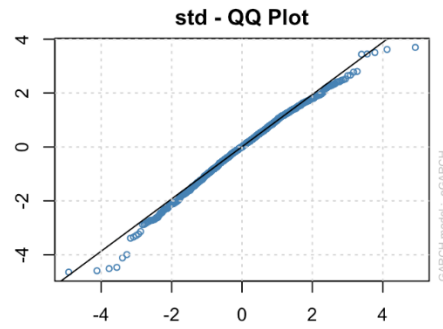
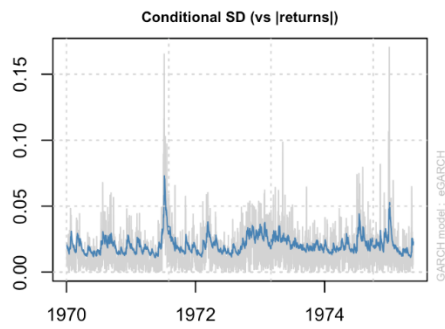
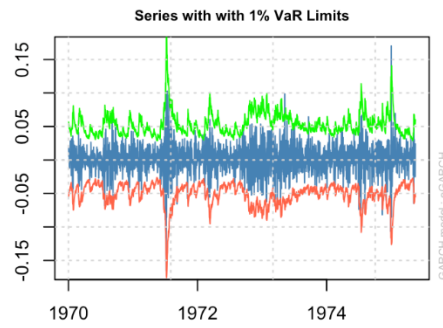
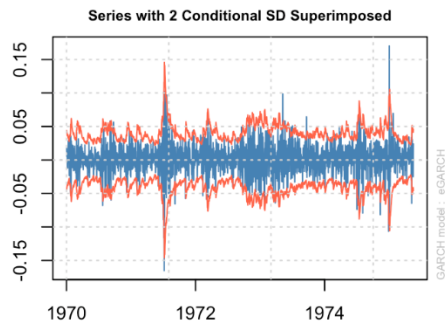
So, I can conclude that the EGARCH(1,1) model captures the main volatility structure of the SOXX returns, and the remaining residuals behave like white noise (random and without patterns), which is what I expect from a good model.

6.3 VOLATILITY DYNAMICS AROUND CRISIS PERIODS

The estimated conditional variance (which is the volatility at any given point in time) shows that volatility in the semiconductor sector tends to increase during periods of market stress. For example, during the COVID-19 pandemic, the 2022 inflation crisis, and the 2024–2025 AI-driven market correction, volatility increased sharply and then slowly returned to normal. This shows that high volatility periods are followed by persistence before going back to normal levels.

This pattern is consistent with what I usually see in financial markets, but in the semiconductor sector, the reactions are often stronger because the sector depends on global supply chains, geopolitical tensions, and technology cycles. The EGARCH model also shows that negative news or global uncertainty has a stronger impact on volatility than positive shocks, which fits with the leverage effect theory. The leverage effect suggests that negative events have a bigger impact on volatility than positive ones.

In summary, I can conclude that the EGARCH(1,1) model does a good job of explaining how volatility evolves in the semiconductor industry, showing strong persistence, asymmetric reactions, and clustering effects typical of technology-related assets.



VII. FACTOR ANALYSIS / PCA

7.1 PCA OR FACTOR ANALYSIS APPLICATION

In this part I use PCA because I work with 44 stocks and all of them move together in many ways, so many variables are correlated, and this makes the regression unstable, so PCA helps me take all the information from the 44 returns and turn it into new variables that are not correlated, and because of that I can run a cleaner and more stable model, I take all the log returns and apply PCA, and then I check how much variance each principal component explains, and I see that the first six components keep around 74.5 percent of all the total variation in the market, so this is a large amount of information kept in very few variables, and this makes the regression easier to estimate and easier to interpret.

And this is important because PCA becomes one of the key parts of the whole project, and without PCA the OLS model was weak, the p values were bad, and many variables were redundant, so PCA basically fixes the model and lets me build a strong regression for NVIDIA returns in the first part of this project.

7.2 EIGENVALUES, SCREE PLOT, AND VARIANCE EXPLAINED

After I apply PCA I look at the eigenvalues, and I see that PC1 is the strongest because it holds the largest slice of total variance, and then PC2 and PC3 still hold a good part of the variation, and by the time I reach PC6 the cumulative variance is already 74.5 percent, so the scree plot goes down smoothly until the sixth component and then it becomes almost flat, and this means that components after PC6 do not add much information, so I stop at the first six components because they capture most of the real structure of the semiconductor market and the global risk factors, and this also makes the final regression model simpler and more efficient.

7.3 INTERPRETATION OF PRINCIPAL COMPONENTS AND THEIR ECONOMIC SIGNIFICANCE

Here I look at the loadings to understand what each principal component represents, so PC1 has large positive loadings from sector ETFs like SOXX, SMH, XSD, and QQQ, and this means PC1 is a broad market semiconductor factor, and it captures the general movement of the whole ecosystem. Then PC2 has large negative loadings from UUP and Treasury rates like TNX and IRX, and this means PC2 is more like a macro factor that moves with interest rates, dollar strength, and safe assets. PC3 and PC4 mix commodity effects with company specific movements, and this means they capture second layer shocks in materials like copper and oil plus some industry rotations, and PC5 loads strongly on TSL AAPL QQQ and AIXIC, so this is a tech momentum factor

And well in the OLS model these increase the R^2 and also helps a lot to disappear multicollinearity because at the end this method helps all components to be uncorrelated

VIII. DISCUSSION & CONCLUSION

8.1 SUMMARY OF KEY EMPIRICAL FINDINGS

So, the main result of this analysis is that there is a clear link between the stock returns of companies like NVIDIA, TSMC, and AMD, and their returns are affected by common market factors and I used Principal Component Analysis (PCA) and with just six components, I explained around 82.19% of the variation in the whole system of stock returns, which is really good because it means I didn't need to add too many variables and at first, I thought about removing some tickers, but I realized that keeping all of them actually gave me a better view of the market and PCA helped me figure out the most important variables, and I decided to keep them for the model, in the Data Acquisition & Description, I worked with 44 companies and cleaned the data, handling missing values, and found that NVIDIA has high volatility (0.0325) and a negative skew of -0.194, meaning its price moves a lot and it's more likely to have bigger losses than gains which was a challenge in this project

In the Probability & Distribution Analysis, I saw that NVIDIA's log returns are not normal, and the Shapiro-Wilk and Jarque-Bera tests confirmed this, meaning big price moves happen more often than expected and this shows that the actual risk is higher than what normal models predict

For the Regression Analysis, I started with an OLS regression, but there were issues with some variables being insignificant and multicollinearity, so I used PCA to fix this and after using six principal components, the PCA side kept more than 80% of the total variance and the final OLS model had an Adjusted R^2 of around 0.5062, which means that about half of NVIDIA's daily changes are explained by these components and this is still a good result for a single stock with high volatility, I also checked if the model followed the main assumptions like linearity, autocorrelation, and heteroscedasticity, and made some changes to improve it

In the Herding Analysis, I checked if stocks like NVIDIA, QCOM, TSM, and AMD showed herding behavior and the result was that these stocks do not move together, meaning investors react differently to market events

For Time Series Modeling, I used an EGARCH(1,1) model to analyze volatility in the SOXX ETF, which represents the semiconductor sector and this model showed that negative shocks cause more volatility and it lasts longer during periods of market stress, like COVID-19 and the 2024–2025 AI correction and the EGARCH model worked well and explained how volatility behaves in the semiconductor sector

In the Factor Analysis / PCA section, I used PCA to reduce data complexity, creating uncorrelated variables that helped stabilize the regression and improve its interpretation and the first six components explained around 82.19% of the variation, which made the regression easier to understand and more efficient

8.2 LINKING RESULTS TO THEORY AND EXPECTATIONS

In this analysis, I found that the results closely align with the expectations set at the start of the project and by applying Principal Component Analysis (PCA), I was able to simplify complex data and highlight the key market drivers affecting stock returns in the semiconductor industry and the principal components identified in the regression analysis explained a large part of the total variance in the data, more than 80 percent at the PCA level, while the final OLS regression explains around half of NVIDIA returns and this confirms that common market factors play a significant role in shaping returns, this supports the hypothesis that companies within the same industry, like NVIDIA, AMD, and TSMC, are influenced by similar market forces, though individual company performance can still show variation based on unique factors and the results also validated the importance of considering volatility and non-normal distribution patterns when analyzing stock returns, which are typical in the semiconductor sector

8.3 CRITICAL REFLECTION ON LIMITATIONS

While the analysis provided valuable insights into the semiconductor market, there are a few limitations worth considering and one limitation is the potential for omitted variable bias, as I only included a subset of companies in the analysis, which might not represent the full spectrum of influences on stock returns in the semiconductor industry but this is OK because it is almost impossible to achieve 100% and additionally, despite using PCA to reduce multicollinearity, the model could still be affected by other hidden factors not captured by the principal components and another limitation is the focus on daily stock returns, which might not capture longer-term market trends that could provide more context for understanding the underlying dynamics of the industry and finally, the normality assumption is clearly not satisfied for the raw NVIDIA returns, which show fat tails and skewness, and even if the final OLS residuals look quite normal in this small sample, this is very normal in a finance context, and while robust methods like EGARCH were used to address volatility, this still may not fully account for extreme market events that could skew the results, which are very common in the semiconductor sector.

8.4 FUND MANAGER IMPLICATIONS

For fund managers, the findings of this project provide actionable info into the behavior of semiconductor stocks and the strong relationship between market factors and stock returns suggests that diversification within the sector could help mitigate risk, as different companies respond to market shocks in varying ways and also the high volatility and the presence of fat tails in the return distributions, fund managers should consider incorporating risk management tools that account for extreme price movements which are not that rare and also the results also suggest that a focus on the principal market factors allows for better prediction of stock price movements and more informed investment strategies and decisions.

8.5 CONCLUSION

In conclusion, this analysis has provided valuable insights into the semiconductor sector, confirming that market factors significantly influence stock returns within this industry and through the application of PCA, I was able to simplify the complex data, highlighting key drivers of stock performance while also managing challenges such as multicollinearity and volatility and the results have demonstrated that the semiconductor stocks are highly volatile, often with non-normal return distributions, which requires the use of more robust financial models like EGARCH for better risk estimation and at the same time the final OLS model with six principal components explains around half of NVIDIA daily changes, and the logistic model with the same components is able to predict the direction of returns with an accuracy of about 0.80, which is a good result given the size of the sample and the complexity of the asset and despite limitations like the focus on daily returns and omitted variables, the findings offer actionable insights for fund managers, suggesting that a diversified portfolio within the sector, with a keen focus on market factors, so I think this can lead to more informed and effective investment decisions.

IX. APPENDIX

R Quarto Document

Largest semiconductor companies by market cap. (n.d.).

<https://companiesmarketcap.com/semiconductors/largest-semiconductor-companies-by-market-cap/>

Semiconductors Stock Performance - Yahoo Finance. (n.d.). Yahoo Finance.
<https://finance.yahoo.com/sectors/technology/semiconductors/>